

Falak Rana

+91 9725556991 | ranafalak18@gmail.com | linkedin.com/in/falak-rana | github.com/falakrana | falakrana.in

TECHNICAL SKILLS

Programming: Python, JavaScript, TypeScript, Rust, Java, C#

Frontend: React.js, Next.js, Angular, HTML5, CSS3, Tailwind CSS

Backend: FastAPI, .NET, Flask, Springboot, Node.js, Express.js, REST APIs, Microservices Architecture

AI / ML: Supervised Learning, Deep Learning, PyTorch, TensorFlow, LangChain, Retrieval-Augmented Generation (RAG)

Databases: MongoDB, MySQL, PostgreSQL, SQLite, Redis

DevOps & Tools: Git, GitHub, Docker, Linux, Postman, AWS (EC2, S3, CloudWatch)

EXPERIENCE

Software Development Intern

Jan 2026 – May 2026

Infodesk India Private Limited (Onsite)

Developing scalable microservices and REST APIs using **.NET (C#)** and **Spring Boot (Java)** for enterprise systems

Building responsive UIs with **React**, **Next.js**, **TypeScript** and managing async workflows using **Redux Saga**

Managing cloud infrastructure on **AWS** (EC2, S3, CloudWatch) for automation, logging, and deployment

Designing and optimizing databases using **SQL** and **MongoDB**

Developing high-performance **Rust** CLI tools for legal document analysis

Machine Learning Intern

Jan 2025 – Apr 2025

UnifiedMentor (Remote)

Delivered 6 client-ready ML applications including prediction systems and **AI APIs**.

Built **Flask-based AI services** to expose trained models for real-world use.

Optimized models for better accuracy and faster inference.

PROJECTS

CacheFlux

Python, FastAPI, Redis, PostgreSQL, Docker

Designed a distributed caching layer using Redis over PostgreSQL, significantly improving API response time under high-traffic workloads.

Implemented cache-aside pattern with TTL and invalidation strategies; containerized using Docker for scalable deployment.

Menova

Python, FastAPI, MongoDB, React, Clerk, Cloudflare R2

Built a multi-tenant SaaS platform for restaurant menus, featuring a real-time branding engine with device simulation and a secure multi-source image upload pipeline (Camera/URL/Local).

Architected a dual-layer order management system using MongoDB TTL indexes for performance-optimized tracking and integrated business analytics via complex aggregation pipelines.

Coursera Main Page Assistant

Python, RAG, Playwright

Developed a browser extension leveraging Retrieval-Augmented Generation (RAG) to answer course-specific queries contextually.

Integrated Gemini API for real-time responses, enhancing learning experience with accurate and contextual assistance.

EDUCATION

Parul University

B.Tech in Computer Science Engineering

2022 – 2026

CGPA: 8.34/10.00

Green Valley High School

12th Grade

2022

91.80%

CERTIFICATIONS

• **Data Science Bootcamp** (Udemy)

• **Linux Command Line & Shell Scripting** (Cisco)

• **Tableau for Data Visualization** (Udemy)

• **AWS Cloud Practitioner** (AWS Cloud)